

The Khronon Route: the Blanchet–Skordis Theory Through Two Gates, and Why No Kinetic Function Repairs It

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Abstract

Version 2 corrects version 1. Version 1 presented a “cubic, one-scale khronon” as a repair that passes cosmology and KiDS. It does not. At the parameters that passed cosmology, the cubic term dominates the static equation in galaxies, and MOND ends at 1–4 kpc. The repair is withdrawn, and a general identity explains why no kinetic function can supply one. PAPER32 named one open door: a khronon sector that is itself the dark fluid, as in the theory of Blanchet & Skordis (JCAP 11 (2024) 040). We run that theory through the same two gates, using its own equations. *Moving source (passed).* Its phantom is a conserved fluid carried by the aether, and a galaxy moving uniformly with its phantom is an exact post-Newtonian solution. The costs: the phantom is assembled rather than generated, it follows a change of velocity only inside $r \approx 2GM_b/\Delta u^2$, and the static halo is stable only for a monotone phantom law. *KiDS external field and σ_8 (failed).* The MOND function reads the gradient of the khronon’s own potential. The benchmark $\mu^{-1} = 22.3 \text{ Mpc}$ puts 24 times the KiDS-1000 bound on it. The benchmark 223 kpc passes, but its mass term ends MOND-like lensing at $\sim 80 \text{ kpc}$. The theory’s own scale-dependent sound speed gives linear $\sigma_8 \approx 0.21\text{--}0.26$. *The identity.* For any kinetic function $\mathcal{K}(\mathcal{Q})$, the static mass term a galaxy feels and the khronon’s late cosmological Jeans scale are the same scale: $\mu_{\text{eff}}^2 = k_J^2(1 + c_{\text{ad}}^2)$, exactly. KiDS’s field bound needs that scale below 0.22 Mpc, while MOND-like lensing to 1 Mpc needs it above $\sim 10 \text{ Mpc}$. The khronon-as-dark-fluid aether is therefore closed by KiDS for every $\mathcal{K}$. *The mass budget.* Even if galaxies accrete the dark fluid as ΛCDM halos do, the conserved phantom of a KiDS lens runs out at 0.13–0.67 Mpc, inside the clean 0.3 Mpc range for most lens masses. *What stands.* A one-scale MOND function $\mathcal{J} = \Lambda j(\mathcal{Y}/\Lambda)$ makes $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ an identity of the action, with $\kappa = 2\sqrt{8\pi}/(3\beta)$; this is algebra, and κ is not derived. The design lesson is that the MOND function must not read the acceleration of the cosmological dark fluid.

1 The route

PAPER32 [Z26d] ended with a specification any completion of $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ must meet. It named as an open door “a khronon sector that itself behaves as dust on FRW, the route of Blanchet & Skordis”. That theory [BS24] is

$$S = \frac{c^3}{16\pi G} \int \sqrt{-g} [R - 2\mathcal{J}(\mathcal{Y}) + 2\mathcal{K}(\mathcal{Q})] + S_m, \quad \mathcal{Q} = c\sqrt{-g^{\mu\nu}\partial_\mu\tau\partial_\nu\tau}, \quad \mathcal{Y} = \frac{A_\mu A^\mu}{c^4}, \quad (1)$$

where $A_\mu = c^2 n^\nu \nabla_\nu n_\mu$ is the acceleration of the khronon’s unit normal n_μ . Their MOND function is $\mathcal{J} = \Lambda - \mathcal{Y} + \frac{2c^2}{3a_0}\mathcal{Y}^{3/2} + \dots$ at low acceleration (their eq. 37), with $\mathcal{J} \rightarrow \Lambda_\infty$ at high acceleration (eq. 40). Their kinetic function is $\mathcal{K} = \mu^2(\mathcal{Q} - 1)^2 + \dots$ (eq. 7), in a DBI form (eq. 87) for cosmology. Their post-Newtonian system (eqs. 19–30) has three parts:

- the unitary-gauge lapse $\Xi = \phi - \dot{\sigma} + \frac{1}{2}|\nabla\sigma|^2$, with $\tau = t + \sigma/c^2$;
- a khronon mass density $4\pi G\rho_\tau = -[\nabla \cdot (\mathcal{J}_\mathcal{Y}\nabla\Xi) + \mu^2\Xi]$, which is the MOND phantom;
- a khronon velocity $v_\tau = -\nabla\sigma$.

In the static case this reduces to AQUAL with a mass term (eq. 35):

$$\nabla \cdot [f(|\nabla\phi|/a_0)\nabla\phi] + \mu^2\phi = 4\pi G\rho_m, \quad f = 1 + \mathcal{J}_\mathcal{Y}. \quad (2)$$

All numbers below come from committed scripts (lanes BSK1, BSX1, BSX2, BSX3) run on both a_0 footings where a_0 enters. Each lane has a mutation control that must fail.

2 Gate 1: the moving source — passed (BSK1, 8/8)

In unitary gauge the extra sector depends only on the lapse and its spatial gradient: $\mathcal{Q} = 1/N$ and $\mathcal{Y} = \gamma^{ij}\partial_i \ln N \partial_j \ln N$. That is the shift-free class in which L330 found a frozen phantom. BS24’s own equations show why it is not frozen. The divergence of the momentum equation, combined with the time derivative of the Poisson equation and matter continuity, gives

$$\partial_t \rho_\tau + \nabla \cdot (\rho_\tau v_\tau) = 0.$$

The phantom is therefore a conserved fluid carried by the aether’s flow. About flat space its transport is second order: that is L330’s identity, and BS24’s dispersion $\omega = 0$. About a galaxy it is first order. A galaxy moving uniformly with its phantom is an exact solution. With $\sigma = -V \cdot x + V^2 t/2$ the aether moves with the galaxy, $\Xi = \phi$, and eqs. 27a, 29a and 35 all hold (symbolic, in comoving coordinates). With the aether pinned (the control) the same moving phantom violates continuity. Passing the gate has three costs:

- (i) The phantom is conserved, so it is assembled from the cosmic khronon dust, never generated by baryons.
- (ii) About a static halo the fluid obeys, in the WKB approximation,

$$\omega^2 = \frac{4\pi G \rho_\tau (k^2 f_\theta - \mu^2)}{k^2(1 - f_\theta) + \mu^2}, \quad f_\theta = f + y f'(y) \cos^2 \theta. \quad (3)$$

Its static limit is AQUAL’s linear response. For $k \gg \mu$ the frequency is k -independent, so the phantom follows a change of velocity Δu only inside $r \approx 2GM_b/\Delta u^2$: 1.3 kpc for the Milky Way at 620 km s^{-1} , 860 kpc for a $10^{14} M_\odot$ cluster at 1000 km s^{-1} .

- (iii) The halo is stable only if $0 < f_\theta < 1$, i.e. only if the phantom acceleration increases with g_N . This is the monotone-phantom condition L340 found, reached here from the fluid side. Any tail that recovers GR (their eq. 40) makes the phantom fall. Where it falls the fluid is unstable at rate² $= (2 + 2/n)GM/r^3$, the local orbital rate, whatever the tail’s amplitude. With ν_{RAR} the unstable region is the inner Milky Way, with e-folding times of 1–4 Myr at 1–3 kpc. Placing the fall above $g \sim 10^3 a_0$ confines it to where the phantom is below 10^{-3} of the field.

3 Gate 2 and σ_8 : the theory as built fails (BSX1, 8/8)

The argument of the MOND function is the aether’s acceleration, $A = \nabla \Upsilon$, with $\Upsilon = -4\pi G a^2 \rho_K \Delta_K / (k^2 + k_J^2)$ (their eqs. 61b and 67). This is the khronon’s own potential gradient, filtered above the late Jeans length. The late Jeans wavenumber is $k_J = a\mu$ exactly, as the check confirms. The field is not the web’s Newtonian field; that is the field which closed the programme’s C-H/K construction [Z26d]. We integrate two-fluid growth (baryons plus the khronon, with their DBI $w(a)$, $c_{\text{ad}}^2(a)$ and the scale-dependent sound speed of their eq. 71) from $z = 1000$. The controls reproduce the Λ CDM $\sigma_8 = 0.811$, and the Λ CDM web field at $z = 0.25$ comes out at $0.0134 a_0$ against an independent mock’s 0.0150. A pure-dust khronon returns $\sigma_8 = 0.82\text{--}0.85$, so the suppression below is the sound speed alone.

benchmark (their Fig. 1)	MOND-function field at KiDS lenses	vs bound 7.2×10^{-5}	r_C (Milky Way)	linear σ_8
A: $\mu^{-1} = 22.3 \text{ Mpc}$, $\lambda_D = 1$	$1.7 \times 10^{-3} a_0$	$\times 24$, fails	1.7 Mpc	0.21
B: $\mu^{-1} = 223 \text{ kpc}$, $\lambda_D = 30$	$4.0 \times 10^{-5} a_0$	$\times 0.55$, passes	0.08 Mpc	0.26

Across $\mu^{-1} = 0.03\text{--}100 \text{ Mpc}$ (with $\lambda_D = 1$ and 30), KiDS alone leaves no μ . The external-field bound needs $\mu^{-1} \lesssim 0.3 \text{ Mpc}$, while MOND-like lensing to $\sim 1 \text{ Mpc}$ needs $r_C = (r_M/\mu^2)^{1/3} \gtrsim 1 \text{ Mpc}$ (their eq. 38), i.e. $\mu^{-1} \gtrsim 10 \text{ Mpc}$. At the same time σ_8 lies between 0.16 and 0.49 at every scanned point. Misteale, McGaugh & Hossienfelder [MMH23] confronted the full static profile of AeST’s identical mass term with KiDS. BS24 checked $w(a)$ against GDM constraints and left the scale-dependent sound speed “for a future investigation”; it is integrated here. Swarm lanes in the same folder report the same suppression on a Λ CDM background (BSK2) and a KiDS failure of khronon dust at the cosmic abundance (BSK3, $\Delta\chi^2 = +374/+364$ on all points). Neither is reproduced here.

4 The identity: one scale for every kinetic function (BSX3, 5/5)

On FRW, BS24’s khronon has $8\pi G \rho = \mathcal{Q} \mathcal{K}_{\mathcal{Q}} - \mathcal{K}$ and $8\pi G P = \mathcal{K}$. Hence $c_{\text{ad}}^2 = \mathcal{K}_{\mathcal{Q}}/(\mathcal{Q} \mathcal{K}_{\mathcal{Q}\mathcal{Q}})$, and the Jeans wavenumber of their eq. 71 is $k_J^2 = 4\pi G \rho(1+w)/c_{\text{ad}}^2 = \frac{1}{2} \mathcal{Q}^2 \mathcal{K}_{\mathcal{Q}\mathcal{Q}}$. In a static region $\mathcal{Q} = \bar{\mathcal{Q}}(1 - \Xi/c^2)$ (their eq. 18a with the cosmic $\bar{\mathcal{Q}}$), so the kinetic part of the 00 equation linearises to $\mu_{\text{eff}}^2 \Xi$ with

$$\mu_{\text{eff}}^2 = \frac{1}{2} \bar{\mathcal{Q}}(\mathcal{K}_{\mathcal{Q}} + \bar{\mathcal{Q}} \mathcal{K}_{\mathcal{Q}\mathcal{Q}}) \quad \implies \quad \frac{\mu_{\text{eff}}^2}{k_J^2} = 1 + c_{\text{ad}}^2 \quad (4)$$

exactly (symbolic, general $\mathcal{K}$). BS24’s quadratic $\mathcal{K}$ reproduces their $\mu^2 \Xi$ (eq. 25) and their $k_J = a\mu$. The same second derivative at the cosmic background sets both the range of the galactic mass term and the late Jeans length, so no choice of $\mathcal{K}$ separates them. With Sec. 3 this closes the route. KiDS’s external-field bound needs $1/k_J \lesssim$

0.22 Mpc at $z = 0.25$. MOND-like lensing to 1 Mpc around a Milky-Way lens needs $r_C = (r_M/\mu_{\text{eff}}^2)^{1/3} \geq 1$ Mpc, i.e. $1/\mu_{\text{eff}} \gtrsim 10$ Mpc. The factor- ~ 46 gap is larger than the order-of-magnitude uncertainty in their eq. 38.

5 Version 1’s repair, withdrawn

Version 1 proposed $\mathcal{K} = \frac{2M^2}{3}(\mathcal{Q} - 1)^3$ with no quadratic term. It passed σ_8 , the forest proxy, KiDS and the early equation of state for late Jeans lengths of 0.3–3 kpc (lane BSX2, which stays on the record). It claimed that the cubic term drops out of galaxies because it is formally 1PN. That was a $c \rightarrow \infty$ limit at fixed M^2 , and it does not hold at the parameters used: $M^2 \sim 10^{15}\text{--}10^{17} \text{ Mpc}^{-2}$. Evaluated there (BSX3 I2), the mass-term range equals the Jeans length (0.3–3 kpc). MOND then ends at $r_C = 0.9\text{--}4.4$ kpc for the Milky Way, and the kinetic sector’s density $M^2\Xi^2/(4\pi Gc^2)$ exceeds the MOND phantom by $10^8\text{--}10^{12}$ at 10 kpc. The control reproduces the slip: dropping the cosmic background from $\mathcal{Q}$, as a formal PN count does, makes the identity fail.

6 The mass budget (BSX3)

A conserved phantom must be assembled from dark fluid that fell in. The deep-MOND phantom grows as $M_{\text{ph}}(< r) = (r/r_M - 1)M_b$. We compare it with two budgets:

- a generous one, the Λ CDM halo mass M_{200} of the lens’s stellar mass (Moster, Naab & White 2013 at $z = 0.25$);
- a tight one, the galaxy’s own cosmic share $(\Omega_{\text{dm}}/\Omega_b)M_b$.

For $\log M_* = 10.0\text{--}10.9$ the generous budget runs out at 0.13–0.67 Mpc; in 6 of the 8 lens-mass and footing cases that is inside KiDS’s clean isolated range of 0.3 Mpc. The tight budget runs out by 82 kpc. So a conserved phantom reproduces KiDS’s MOND-like lensing only if galaxies accrete more dark fluid than Λ CDM halos hold at those radii. The swarm lane BSK3’s KiDS failure corresponds to the tight budget; it is not reproduced here.

7 The programme’s relation, inside the action

$\mathcal{J}(0) = \Lambda$ is the cosmological constant that cosmology measures, and the $\mathcal{Y}^{3/2}$ coefficient is $2c^2/(3a_0)$ (their eq. 37). With one scale in $\mathcal{J}$,

$$a_0 = \frac{2c^2\sqrt{\Lambda}}{3\beta} = \kappa c\sqrt{G\rho_\Lambda}, \quad \kappa = \frac{2\sqrt{8\pi}}{3\beta}, \quad \rho_\Lambda = \frac{\Lambda c^2}{8\pi G} \quad (5)$$

(symbolic). So $\kappa = \frac{1}{2}$ is $\beta = 4\sqrt{8\pi}/3 = 6.684$, and Planck’s Λ ($h = 0.6736$, $\Omega_\Lambda = 0.6847$) then returns $a_0 = 9.35 \times 10^{-11} \text{ m s}^{-2}$, the canonical footing. The relation stops being a coincidence between two constants: a_0 and the vacuum energy become two coefficients of one function. The programme’s earlier principle that there is no independent a_0 [Z26b] becomes a property of the action. The coefficient remains a shape number and is not derived, consistent with the zero-mode theorem [Z26a].

8 What this leaves

The khronon route closes on its own equations, for every $\mathcal{K}$. The lesson is structural.

- The MOND function must not read the acceleration of the cosmological dark fluid: its Jeans physics then fixes the galactic mass term.
- It must not read the web’s Newtonian field either, which is what closed C-H/K in PAPER32.

What survives is a completion in which the phantom is generated by the baryons (C-H/K’s λ -channel does this, unlike a conserved fluid), the MOND kernel is blind to the large-scale field, and the dark matter is a separate component invisible to the kernel. That route is being built elsewhere in the programme. The one-scale identity $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ of Sec. 7 is algebra and carries over to any action whose MOND function contains the cosmological constant as its zero-acceleration value. Blanchet & Skordis’s Khronon-Tensor theory [BS25] is not examined.

9 Reproducibility

All lanes are in <https://github.com/carlzimmerman/zimmerman-formula> under `real_research/bs_khronon_2026/`. They run in about a minute each and need no survey data files.

lane	script	checks	mutation control (must fail)
BSK1	<code>BSK1_moving_source_gate.py</code>	8/8	aether pinned: moving solution and fluid response fail
BSX1	<code>BSX1_kids_external_field_gate.py</code>	8/8	khronon made pure dust: the σ_8 and field suppressions vanish
BSX2	<code>BSX2_cubic_one_scale_khronon.py</code>	3/3 ^a	quadratic term restored at the same Jeans length: mass term returns, forest and early w fail
BSX3	<code>BSX3_jeans_mass_identity_and_budget.py</code>	5/5	cosmic background dropped from $\mathcal{Q}$ (v1's slip): identity and I2 fail

^a Its cosmology checks stand; its X1 (galaxies unaffected) is superseded by BSX3 I2.

References

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